

9.8-9.9 Angular momentum

I don't understand the steps/process that went into finding the spherical harmonics (how did we go from a $|l, l\rangle$ to a $|l, m\rangle$, with some extra trig sprinkled in)?

$$\hat{L}_z \rightarrow \frac{\hbar}{i} \frac{\partial}{\partial \phi}$$

vs linear momentum

$$\hat{p}_x \rightarrow \frac{\hbar}{i} \frac{\partial}{\partial x}$$

$$\hat{L}_z |l, m\rangle = m\hbar |l, m\rangle,$$

$$\hat{L}_x \rightarrow \frac{\hbar}{i} \left(-\sin \phi \frac{\partial}{\partial \theta} - \cot \theta \cos \phi \frac{\partial}{\partial \phi} \right)$$

$$\hat{L}_y \rightarrow \frac{\hbar}{i} \left(\cos \phi \frac{\partial}{\partial \theta} - \cot \theta \sin \phi \frac{\partial}{\partial \phi} \right)$$

$$\hat{L}^2 \rightarrow -\hbar^2 \left[\frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial}{\partial \theta} \right) + \frac{1}{\sin^2 \theta} \frac{\partial^2}{\partial \phi^2} \right]$$

$$Y_{l,m}(\theta, \phi) = \frac{(-1)^l}{2^l l!} \sqrt{\frac{(2l+1)(l+m)!}{4\pi(l-m)!}} e^{im\phi} \frac{1}{\sin^m \theta} \frac{d^{l-m}}{d(\cos \theta)^{l-m}} \sin^{2l} \theta \quad (9.148)$$

$$Y_{l,-m}(\theta, \phi) = (-1)^m [Y_{l,m}(\theta, \phi)]^* \quad (9.150)$$

spherical harmonics with $l = 0, 1$, and 2 :

$$Y_{0,0}(\theta, \phi) = \sqrt{\frac{1}{4\pi}} \quad (9.151)$$

$$Y_{1,\pm 1}(\theta, \phi) = \mp \sqrt{\frac{3}{8\pi}} e^{\pm i\phi} \sin \theta \quad (9.152a)$$

$$Y_{1,0}(\theta, \phi) = \sqrt{\frac{3}{4\pi}} \cos \theta \quad (9.152b)$$

$$Y_{2,\pm 2}(\theta, \phi) = \sqrt{\frac{15}{32\pi}} e^{\pm 2i\phi} \sin^2 \theta \quad (9.153a)$$

$$Y_{2,\pm 1}(\theta, \phi) = \mp \sqrt{\frac{15}{8\pi}} e^{\pm i\phi} \sin \theta \cos \theta \quad (9.153b)$$

$$Y_{2,0}(\theta, \phi) = \sqrt{\frac{5}{16\pi}} (3 \cos^2 \theta - 1) \quad (9.153c)$$